3D Seismic Tomography with Simultaneous Hypocenter Estimation via Expectation–Maximization: Method and Application

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November 1, 2026

Abstract

Keywords: seismic tomography; hypocenter estimation; expectation–maximization; local earthquake tomography; 3D velocity model

1 Introduction

Local-earthquake travel-time tomography uses picked seismic arrival times to estimate three-dimensional velocity structure and earthquake locations. Early work already formulated the inversion in terms of both source and medium parameters (Aki & Lee, 1976). An earthquake hypocenter is not an independently measured datum: its location is inferred from arrival times using an assumed velocity model, while velocity reconstruction in turn depends on the event locations (Kissling, 1988; Thurber, 1992). Treating catalog hypocenters as fixed inputs can therefore propagate location errors and dependencies on the models and procedures used to construct the catalog, particularly when those assumptions or location uncertainties are not documented. This coupling motivates estimating earthquake locations and velocity structure together.

Travel-time tomography remains useful for constructing and refining local velocity models (Fichtner et al., 2024). Such models can also provide starting points for waveform-based inversion, whose results may depend on the initial model (Virieux & Operto, 2009). There is therefore continued demand for practical methods that use arrival-time data to estimate three-dimensional structure without assuming accurately known earthquake locations.

Established joint-inversion methods iteratively update the velocity structure and an individual location estimate for each earthquake (Thurber, 1983). They differ in how those estimates are initialized: tomoDD uses initial event locations (Zhang & Thurber, 2003), whereas LOTOS determines them from arrival times and station coordinates in a one-dimensional reference model before three-dimensional inversion (Koulakov, 2009). In these formulations, locations enter velocity updates as individual estimates rather than as weighted alternatives. Probabilistic methods address different parts of this problem. Event-location methods can characterize alternative hypocenters given a prescribed velocity model (Lomax et al., 2000; Smith et al., 2022). Bayesian joint inversion has been demonstrated for both one-dimensional velocity structure (Ryberg & Haberland, 2019) and three-dimensional local-earthquake tomography using trans-dimensional Monte Carlo sampling (Piana Agostinetti et al., 2015); the latter jointly samples hypocenters and velocity structure without requiring catalog locations as starting coordinates. DeepGEM proposes generalized expectation–maximization for blind inversion (Gao et al., 2021),

while a recent three-dimensional variational double-difference method quantifies uncertainty in the velocity structure (Yang et al., 2025). These approaches differ in their treatment of location uncertainty and computational strategy; joint probabilistic 3D inversion without catalog starting locations is therefore not unique to EMTomo.

A practical motivation is to reconstruct three-dimensional velocity structure when reliable catalog hypocenters are unavailable, without committing the velocity update to a single estimated location for each event. We present EMTomo, which starts from arrival-time picks, station coordinates, and an initial velocity model, but does not require initial hypocenter coordinates. It alternates between evaluating possible event locations in the current model and updating the velocity model using either one selected location (hard assignment) or several weighted location hypotheses (soft assignment). This is a computationally different choice from the joint sampling of hypocenters and velocity structure by Piana Agostinetti et al. (2015); we do not claim that joint probabilistic 3D inversion without initial catalog locations is new. Our question is whether retaining several plausible locations is useful in this iterative travel-time inversion setting, and how the resulting reconstructions compare with an established joint-inversion workflow.

We investigate this question in three synthetic experiment series. First, checkerboard tests keep the inversion-cell size fixed while varying the size of the cubic velocity anomalies; they compare hard-assignment EMTomo with tomoDD initialized from hypocenters perturbed by location noise. Second, a layered velocity model with spherical anomalies, inverted from a layered starting model, is tested with an artificial event–station geometry using hard-assignment EMTomo and tomoDD with and without noise in its initial hypocenters; its matched soft-assignment comparison remains to be completed. Third, the same layered model is tested with event and station locations taken from the 2011 Geysers configuration, including a separate EMTomo soft-assignment candidate-count study. EMTomo requires no initial hypocenters, whereas tomoDD does; the methods also use different velocity parameterizations. We therefore evaluate tomoDD against a trilinear oracle fitted to the known velocity field as a reference for what its representation can resolve, rather than interpreting every difference from the true field as an inversion error. The comparisons characterize these specified setups, not a universal advantage of either method.

[TODO: Before submission: provide a stable repository link and a code-availability statement if the implementation is released. In Discussion, note that joint P/S inversion and time-lapse applications are outside the scope of this study. Remaining experiment-specific checks are listed at the end of Section 3.]

2 Method

This section describes EMTomo's v1.4 framework used in the layered-model experiments. The older checkerboard runs are identified separately in Section 4.1.

2.1 Parameterization and observations

We represent the isotropic velocity model by L values $v_\ell > 0$ at the centers of uniform cubic inversion cells, and perform the update in slowness $s_\ell = 1/v_\ell$. The vector $\mathbf{s} = (s_1, \ldots, s_L)^\mathsf{T}$ contains the slowness values of all inversion cells. Similar blockwise slowness models were used by Aki and Lee (1976). For travel-time calculations, each inversion cell is divided into equal cubic subcells; the subdivision factor sets their number along each side. This does not add inversion parameters. The true model used to generate synthetic data may have an even finer grid. Slowness on the computational grid can be assigned by nearest-neighbor or trilinear interpolation. We use nearest-neighbor interpolation in all experiments: each subcell has the slowness of its parent inversion cell. An initial velocity model is required.

For a catalog of E events, let t_{en} be the observed arrival time of one selected seismic phase from event e at station n. Its predicted value is $o_e + T_n(\mathbf{h}_e; \mathbf{s})$, where $\mathbf{h}_e \in \mathbb{R}^3$ is the hypocenter, o_e is the origin-time offset, and T_n is the travel time to station n. With Gaussian errors, origin time can be integrated out analytically, as in NonLinLoc (Lomax et al., 2000). Given the velocity model and the hypocenter, its estimate is

$$\widehat{o}_e = \frac{\sum_n p_{en}\left[t_{en} - T_n(\mathbf{h}_e; \mathbf{s})\right]}{\sum_n p_{en}}, \qquad p_{en} = \sigma_{en}^{-2}, \tag{1}$$

where σ_{en}^2 is the timing-error variance defined in Section 2.3. This estimate uses the same time reference as the observed arrivals. After eliminating origin times, the problem has $L + 3E$

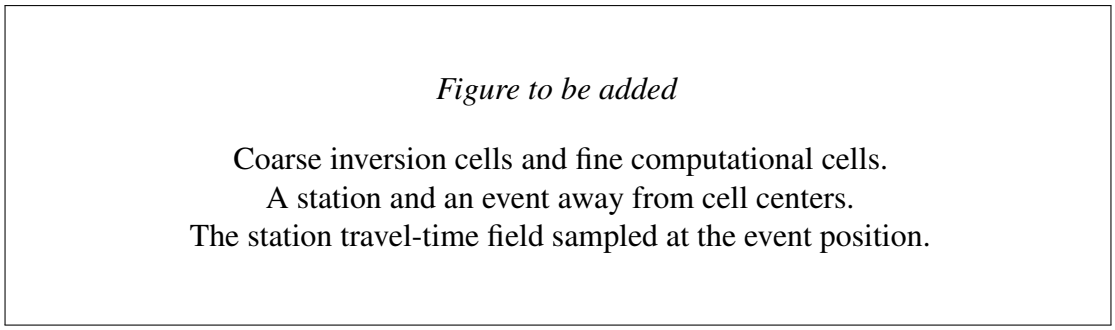


Figure 1: Inversion and computational grids, with travel-time evaluation for one station–event pair.

remaining unknowns: L slowness values and three coordinates per event. We do not require initial hypocenters, but we do need to know which event each arrival belongs to.

2.2 Travel-time fields on the fine grid

On the computational grid defined above, we use the second-order fast-marching solver in `scikit-fmm` (The scikit-fmm team, n.d.) to compute one first-arrival travel-time field $T_n(\mathbf{x}; \mathbf{s})$ per station at each iteration. By reciprocity, the travel time from a station to an event equals the time from the event to the station. We can therefore reuse each station field for all events.

Stations and events need not lie at grid centers. The field initialization uses the actual station coordinates, and trilinear interpolation gives travel times at continuous event coordinates. This interpolates travel times, not the blockwise slowness model. Figure 1 illustrates the grids and the travel-time calculation for one station–event pair.

2.3 Event-location step: candidate positions and weights

We keep the velocity model fixed and use the station travel-time fields computed in the previous step (Section 2.2). Each field gives the travel time between a station and each fine-grid cell center. For a trial event position $\mathbf{h}$, subtracting this travel time from the observed arrival gives a station-specific estimate of origin time:

$$\widehat{o}_{en}(\mathbf{h}) = t_{en} - T_n(\mathbf{h}; \mathbf{s}). \tag{2}$$

We do this for all stations that recorded the event. These estimates refer to the same event time, so we compare them with their weighted mean $\widehat{o}_e(\mathbf{h})$ from Eq. (1). We define the timing residual and the weighted squared misfit as

$$\delta t_{en}(\mathbf{h}) = \widehat{o}_{en}(\mathbf{h}) - \widehat{o}_e(\mathbf{h}), \qquad \chi^2_e(\mathbf{h}) = \sum_n p_{en}(\mathbf{h})\,[\delta t_{en}(\mathbf{h})]^2. \tag{3}$$

A smaller misfit means that the stations give more consistent estimates of origin time. The weights $p_{en} = 1/\sigma^2_{en}$ account for arrival-time uncertainty. We use

$$\sigma^2_{en}(\mathbf{h}) = \big(\alpha T_n(\mathbf{h};\mathbf{s})\big)^2 + \sigma^2_{\mathrm{pick}} + \sigma^2_{\mathrm{model}}, \qquad p_{en}(\mathbf{h}) = \sigma^{-2}_{en}(\mathbf{h}), \tag{4}$$

where α is dimensionless, while σ_{pick} and σ_{model} are in seconds. The term αT_n allows larger timing errors for longer travel times; σ_{pick} describes absolute picking error, and σ_{model} allows for modeling error. The errors are assumed Gaussian, but their variances need not be equal. These three parameters are set before inversion, not estimated during it. By default, α and σ_{pick} are read from the synthetic-data noise settings, unless explicitly overridden; σ_{model} is specified separately.

The squared misfit is the data-fit part of our criterion. Integrating out origin time under a uniform prior gives the full negative log likelihood

$$\Phi_e(\mathbf{h}) = \frac{1}{2}\chi^2_e(\mathbf{h}) + \frac{1}{2}\sum_n \log \sigma^2_{en} + \frac{1}{2}\log\left(\sum_n p_{en}\right). \tag{5}$$

up to a constant shared by all trial positions. The logarithmic terms are needed when the variances change with the trial position. If the variances are fixed, ranking positions by Φ_e is equivalent to ranking them by χ^2_e.

Evaluating Φ_e at all fine-grid cell centers gives a cost field for the event. We select a chosen number of low-cost cells, skipping cells too close to those already selected. The minimum separation is set in fine-grid cells using the largest coordinate difference (Chebyshev distance). We then refine the position within each selected cell by a local search for a lower cost, using trilinearly interpolated travel times. This distance constraint applies to the selected cells, not to the final continuous positions. After refinement, we rank the candidates again and keep a chosen

number of the best locations. For these locations $\mathbf{h}_{ej}$ with costs Φ_{ej}, the weights are

$$w_{ej} = \frac{\exp[-(\Phi_{ej} - \min_k \Phi_{ek})/\theta]}{\sum_k \exp[-(\Phi_{ek} - \min_i \Phi_{ei})/\theta]}, \tag{6}$$

where $\theta > 0$ controls how strongly the weights favor the best candidate; $\theta = 1$ gives normalized likelihood weights.

We may test many candidates but keep only some for the velocity update. Hard mode uses only the best one, with weight one. Soft mode uses the retained candidates with the weights above. Testing many candidates is therefore not the same as using soft mode; with one retained candidate, both modes are identical.

2.4 Velocity-update step: ray sensitivities and inversion

At fixed candidate hypocenters, first-order changes in travel time are approximated by $T_n(\mathbf{h}_{ej}; \mathbf{s} + \delta\mathbf{s}) \simeq T_n(\mathbf{h}_{ej}; \mathbf{s}) + \sum_{\ell=1}^{L} G_{en\ell}^{(j)} \delta s_\ell$. The row $G_{en\ell}^{(j)}$ is approximated by the length of the ray from $\mathbf{h}_{ej}$ to station n within inversion cell ℓ. We compute spatial gradients of the station travel-time fields on the fine grid and interpolate them along each ray. Starting at the candidate hypocenter, we follow the negative gradient toward the station, accumulate path lengths in fine cells, and sum them into the coarse inversion cells. Within the small homogeneous sphere used to initialize the station field, path length is assigned to the station cell, consistently with that local approximation. Rays that do not reach the station are excluded from the update.

For each event and retained hypothesis, let $\mathbf{r}_{ej}$ collect observed minus predicted arrivals at stations with successful rays, let G_{ej} collect their sensitivity rows, and set $p_{en} = 1/\sigma_{en}^2(\mathbf{h}_{ej})$. Rather than add origin-time increments as unknowns, we remove their effect with precision-weighted station means:

$$\widetilde{r}_{en}^{(j)} = r_{en}^{(j)} - \frac{\sum_m p_{em} r_{em}^{(j)}}{\sum_m p_{em}}, \qquad \widetilde{G}_{en:}^{(j)} = G_{en:}^{(j)} - \frac{\sum_m p_{em} G_{em:}^{(j)}}{\sum_m p_{em}}. \tag{7}$$

Only valid stations enter these sums. Holding locations and weights fixed during this step, the

linearized, damped least-squares update minimizes

$$\sum_{e,j} w_{ej} \sum_{n} p_{en} \left(\widetilde{r}_{en}^{(j)} - \widetilde{G}_{en:}^{(j)} \delta \mathbf{s} \right)^2 + \delta \mathbf{s}^{\mathsf{T}} D \, \delta \mathbf{s}, \tag{8}$$

where D is a diagonal damping matrix; its entries can be increased for poorly illuminated cells. Accordingly, the implementation accumulates

$$H = \sum_{e,j} w_{ej} \widetilde{G}_{ej}^{\mathsf{T}} P_{ej} \widetilde{G}_{ej}, \qquad \mathbf{b} = \sum_{e,j} w_{ej} \widetilde{G}_{ej}^{\mathsf{T}} P_{ej} \widetilde{\mathbf{r}}_{ej}, \qquad (H + D)\delta \mathbf{s} = \mathbf{b}, \tag{9}$$

with $P_{ej} = \mathrm{diag}(p_{en})$. We update the slowness by $\delta \mathbf{s}$, convert back to velocity, and optionally limit the fractional velocity change in each cell. Repeating the location and velocity steps for a prescribed number of cycles produces the final model.

2.5 Mathematical basis and limitations

Under independent Gaussian pick errors, Eq. (5) follows from marginalizing the origin time. At $\theta = 1$, Eq. (6) gives posterior weights on the retained hypotheses with equal prior probabilities, not a full spatial posterior. Soft assignment averages their quadratic data-fit contributions; hard assignment retains only the best hypothesis. A related approach is DeepGEM (Gao et al., 2021), which uses a normalizing flow to approximate the source posterior in variational EM for blind seismic tomography; here the approximation is a finite set of weighted locations.

For isotropic propagation, small perturbations, and no ray-path switching, Fermat's principle supports the frozen-ray slowness linearization. Weighted centering is an origin-time-only form of parameter elimination, related to the broader source-parameter separation of Pavlis and Booker (1980); it does not project out spatial hypocenter corrections. With locations, weights, and variances fixed, $H \succeq 0$, and strictly positive damping gives $H + D \succ 0$. The unclipped update therefore uniquely minimizes Eq. (8), reducing this quadratic objective relative to a zero update unless that update is already optimal.

This local result does not establish monotonic improvement of the nonlinear likelihood. The candidate set changes between iterations, ray sensitivities are approximate, and model-dependent variances are frozen in the M-step; tempering and step clipping introduce further departures

from exact EM. Thus the standard EM monotonicity guarantee (Dempster et al., 1977) does not directly apply. Regularization stabilizes the update but cannot resolve velocity–location ambiguities unsupported by the data.

3 Experimental setup

3.1 Forward modeling

Synthetic first-arrival times are generated independently of EMTomo using PyKonal 0.4.1 (White et al., 2020). The generator solves the Cartesian eikonal equation from the true event coordinates on a refined nodal grid, initializes off-grid sources with local straight-segment slowness integrals, and samples receivers by interpolating the regular factor $T(\mathbf{x})/\|\mathbf{x}-\mathbf{h}\|$. This differs from EMTomo's station-centered `scikit-fmm` fields, homogeneous source-sphere initialization, and direct interpolation of travel times. Separate solvers, grids, and source treatments reduce numerical inverse crime; both still approximate the same isotropic first-arrival problem, so this is not a test against independent wave physics.

Before referencing each event to its earliest observed arrival, we add independent relative and absolute Gaussian errors,

$$t_{en}^{\text{noisy}} = T_{en}^{\text{true}} + 0.01 T_{en}^{\text{true}} Z_{en}^{(1)} + \sigma_a Z_{en}^{(2)}, \qquad Z_{en}^{(1)}, Z_{en}^{(2)} \sim \mathcal{N}(0,1), \tag{10}$$

where σ_a is the absolute timing-noise standard deviation. Both methods use the same noisy picks; inter-event differential times for tomoDD are formed from these picks without additional waveform cross-correlation observations. Initial hypocenters for tomoDD are obtained by adding independent zero-mean Gaussian perturbations to each true coordinate, with out-of-domain positions rejection-resampled. EMTomo does not require an initial hypocenter catalog.

3.2 Evaluation

Velocity fields. We evaluate the native reconstructed fields at the same physical points $\{\mathbf{x}_q\}_{q=1}^{Q}$, without converting one method to the other's parameterization. EMTomo is piecewise constant;

tomoDD is evaluated by its native trilinear velocity interpolation, including the fixed exterior nodes. We assess whole-volume velocity accuracy using the root-mean-square error,

$$\mathrm{RMSE}_v = \sqrt{\frac{1}{Q}\sum_q [\widehat{v}(\mathbf{x}_q) - v^{\mathrm{true}}(\mathbf{x}_q)]^2}. \tag{11}$$

For each representation, a truth-only oracle fits velocity values to the known field, without using arrivals or estimating hypocenters. For constant cells, the volume squared-error optimum is the arithmetic mean of true velocity within each cell, *not* the inverse of mean slowness. For tomoDD we use the saved least-squares trilinear truth fit with the same free nodes and fixed exterior values as the inversion. These references quantify representational error, not the resolution attainable from the available rays, and are distinct from an inversion with fixed true hypocenters. A whole-volume oracle is not a lower bound for every regional RMSE, and the difference between reconstruction and oracle RMSE is not an additive decomposition of inversion error.

The checkerboard truth is exactly representable by EMTomo's aligned constant cells, so its oracle is the true field. This is *not* true for the layered sphere models on the coarser inversion grid: they require a separate cell-mean oracle. Because the two layered experiments share the same truth and grids, they share both representation oracles and initial native fields.

Hypocenters. We assess hypocenter accuracy using the 90th percentile of the three-dimensional location errors $d_e = \|\widehat{\mathbf{h}}_e - \mathbf{h}_e^{\mathrm{true}}\|$, in meters, with all events weighted equally. For soft EMTomo, we use the hypothesis with the largest weight as the point estimate.

Spherical anomalies. For each of the six spherical anomalies, we additionally evaluate velocity RMSE over its known nonzero support S_k:

$$\mathrm{RMSE}_{v,k} = \sqrt{\frac{1}{Q_k}\sum_{\mathbf{x}_q \in S_k} [\widehat{v}(\mathbf{x}_q) - v^{\mathrm{true}}(\mathbf{x}_q)]^2}, \qquad k = 1, \ldots, 6, \tag{12}$$

where Q_k is the number of evaluation points in S_k. Thus each reconstruction is characterized by one whole-volume RMSE and six individual anomaly RMSE values. The anomaly supports are defined by the true model, without segmenting the reconstructed field.

4 Results

4.1 Checkerboard test

The checkerboard models occupy a $24 \times 12 \times 12$ km domain. Alternating cubes have velocities 4.75 and 5.25 km/s and sides 6, 4, or 3 km, corresponding to $n_x = 4, 6, 8$ cubes along the X axis. All three models share 1200 events and 98 surface stations (Figure 2). Both methods start from a homogeneous 5 km/s velocity model and use 1-km velocity parameterizations. Arrival-time noise combines 1% relative errors with $\sigma_a = 0.01$ s; the initial tomoDD hypocenters are perturbed with a 1-km standard deviation per coordinate. EMTomo uses 12 hard-assignment updates, selecting one of 15 candidates, with $\lambda = 1$ and coverage-damping power zero; tomoDD uses three joint iterations.

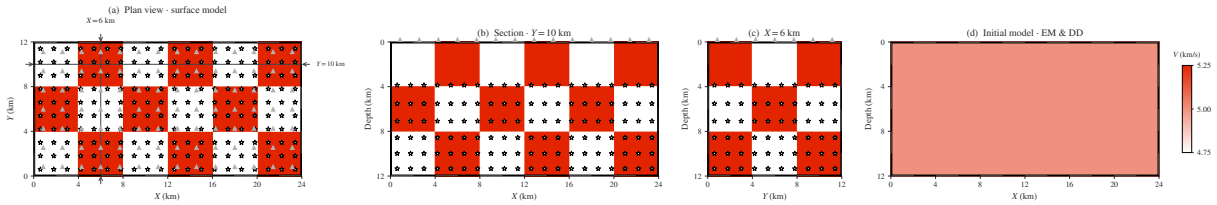


Figure 2: Checkerboard model and acquisition geometry for $n_x = 6$. (a) Surface velocity field and plan-view projections of stations (grey triangles) and events (white stars). (b,c) Vertical sections at $Y = 10$ km and $X = 6$ km, with stations and events projected onto each plane; coincident projections are drawn once. (d) The homogeneous initial velocity model shared by both methods and all three checkerboards. Depth is positive downward.

Figure 3 compares the reconstructed velocity fields with the truth and the trilinear representation oracle. Both methods recover the alternating pattern most clearly in the horizontal sections, while recovery deteriorates at depth. EMTomo retains sharp cell boundaries but shows local variability; tomoDD produces smoother transitions. The trilinear oracle also rounds the true block boundaries, illustrating the contribution of the velocity representation to these differences.

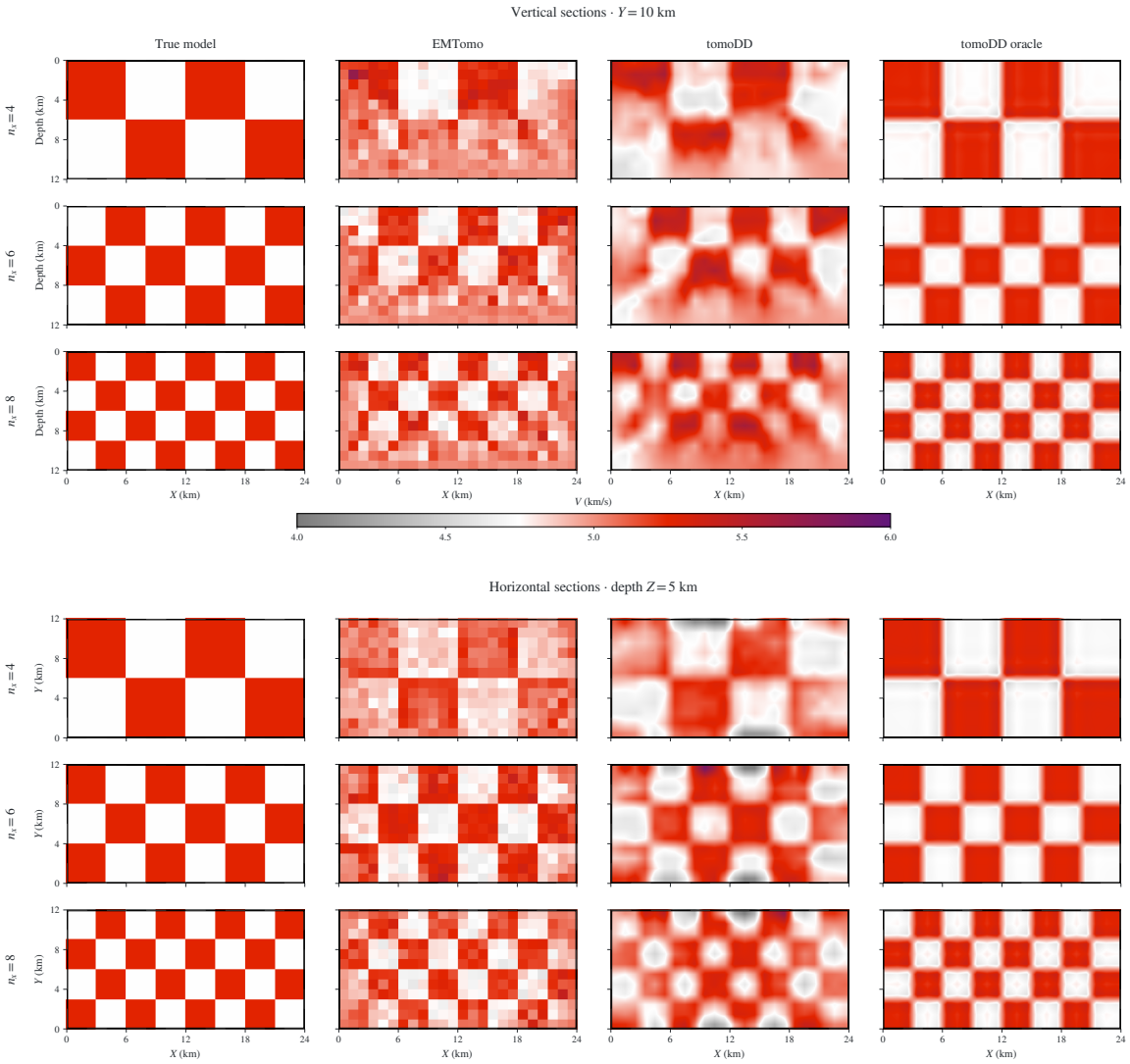


Figure 3: Checkerboard velocity reconstructions. Columns show truth, EMTomo after 12 updates, tomoDD after three iterations, and the trilinear representation oracle. The upper three rows are vertical sections at $Y = 10$ km for $n_x = 4, 6, 8$; the lower three are horizontal sections at $Z = 5$ km in the same order. All panels share the velocity scale in km/s. The oracle is a truth-only representation reference, not an inversion result.

The location comparison for $n_x = 6$ (Figure 4) shows that both methods place events close to their true positions, despite the perturbed starting catalog supplied to tomoDD. Across the full catalogs (Figure 5), the 90th-percentile location errors are 506, 373, and 403 m for EMTomo, compared with 599, 588, and 609 m for tomoDD at $n_x = 4, 6, 8$, respectively.

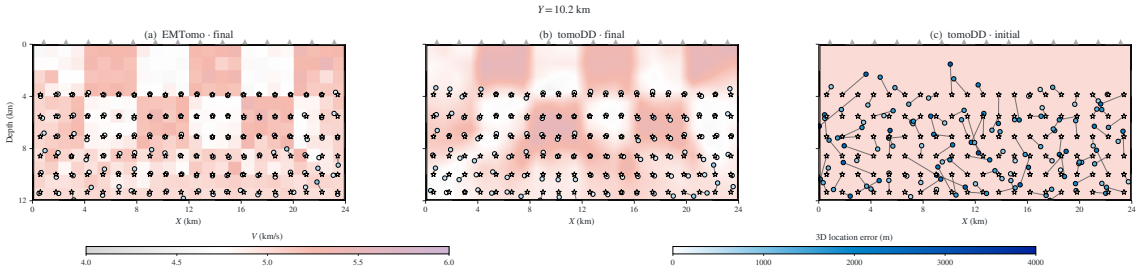


Figure 4: Hypocenter estimates for the 120 events with true $Y = 10.2$ km in the $n_x = 6$ checkerboard: (a) EMTomo, (b) tomoDD, and (c) initial tomoDD locations. White stars mark true positions; circles mark estimates, connected to the truth by grey segments. Positions are projected onto the X–depth plane, while circle colors show full 3D location errors. Backgrounds show the corresponding final or initial velocity fields. EMTomo locations are from the E-step preceding the last velocity update.

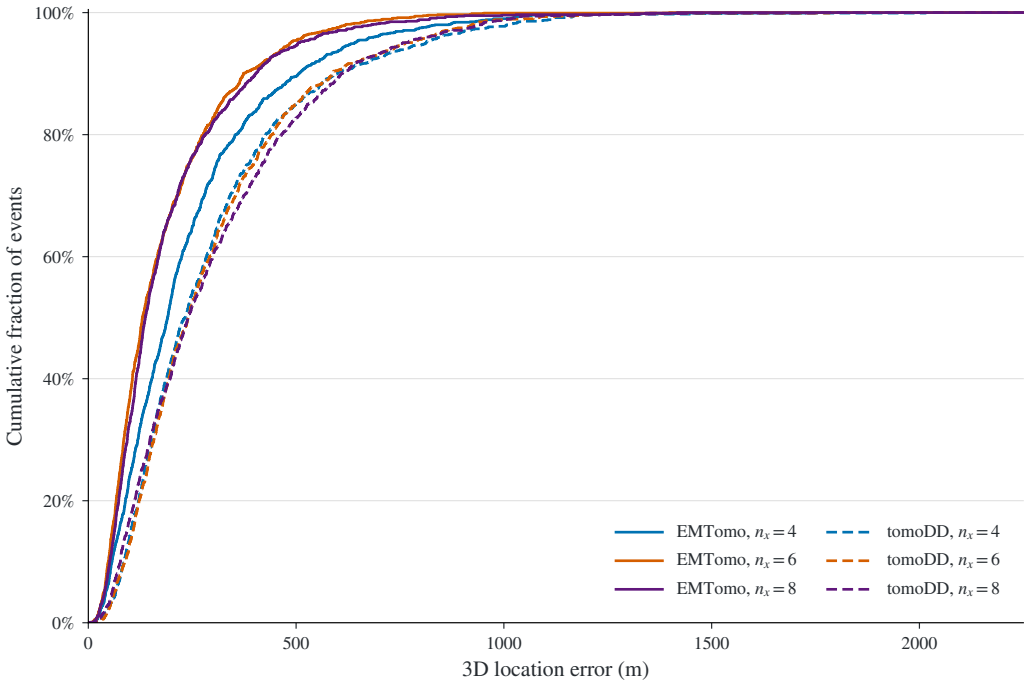


Figure 5: Cumulative distributions of 3D location error for all 1200 events in each checkerboard. Colors identify $n_x = 4, 6, 8$; solid lines denote EMTomo and dashed lines tomoDD. EMTomo uses the E-step locations preceding the twelfth velocity update; tomoDD uses locations after three joint iterations. A higher curve indicates a larger fraction of events within a given error.

4.2 Spherical anomalies

4.3 Real data

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